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Oefeningenreeks 5D nr. 13

$$y^2 = x^2 - x^4$$

$$\Rightarrow 4\int_0^1 \sqrt{x^2 - x^4}dx = 4\int_0^1 x\sqrt{1-x^2}dx$$

$$u = 1 - x^2$$

$$du = -2xdx$$

$$= 4\int_0^1 \frac{-1}{2}\sqrt{u}du = 2\int_0^1 x\sqrt{u}du$$

$$= 2\left[\frac{2u^{3/2}}{3}\right]_0^1$$

$$= \frac{4\cdot 1}{3} - \frac{4\cdot 0}{3}$$

$$= \frac{4}{3}$$

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References